Experiment 21: Factor Analysis and Latent Factor Interpretation

Question

Problem Statement: An HR department runs a 10-question employee-satisfaction survey. Management wants to know which few hidden factors drive the answers. Survey responses from 300 employees on pay, workload, manager support, growth, team spirit and similar items (Likert 1–5). The objective is to identify and interpret latent factors with factor analysis. Use Scikit-learn, NumPy, Pandas, Matplotlib.

Tasks:

1. Load and explore the survey data in Pandas; check correlations.
2. Fit Scikit-learn's FactorAnalysis with 2, 3 and 4 factors.
3. Apply varimax rotation and display the loadings matrix as a heatmap.
4. Name each factor from its high-loading questions (for example, "Compensation", "Work Culture").
5. Compare factor analysis with PCA on the same data.

Code

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"""Experiment 21: Factor Analysis and Latent Factor Interpretation (employee survey)"""
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter
from sklearn.decomposition import FactorAnalysis, PCA
from sklearn.preprocessing import StandardScaler


np.random.seed(42)


# ---------------- Task 1: load / create survey data (300 employees, 10 Likert questions) ----------------
questions = ["Salary_fair", "Benefits", "Bonus_fair", "Manager_support", "Team_spirit",
             "Communication", "Career_growth", "Training", "Promotion_clarity", "Workload_ok"]
theme_of = dict(zip(questions, ["Compensation"] * 3 + ["Work Culture"] * 3 + ["Growth"] * 3 + ["Work
Culture"]))
true_L = np.zeros((10, 3))                                          # hidden structure used to simulate
answers
true_L[0:3, 0] = [0.9, 0.8, 0.85]; true_L[3:6, 1] = [0.85, 0.8, 0.75]
true_L[6:9, 2] = [0.9, 0.8, 0.85]; true_L[9, 1] = 0.5
F = np.random.randn(300, 3)
raw = F @ true_L.T + np.random.normal(0, 0.55, (300, 10))
df = pd.DataFrame(np.clip(np.round(3 + 0.9 * raw), 1, 5).astype(int), columns=questions)   # Likert 1-5
# To use real data:  df = pd.read_csv("survey.csv")
print(df.head()); print("\n", df.describe().round(2).loc[["mean", "std", "min", "max"]])


corr = df.corr()
plt.figure(figsize=(7, 6)); plt.imshow(corr, cmap="coolwarm", vmin=-1, vmax=1); plt.colorbar()
plt.xticks(range(10), questions, rotation=60, ha="right"); plt.yticks(range(10), questions)
plt.title("Correlation between survey questions"); plt.tight_layout(); plt.show()


Xs = StandardScaler().fit_transform(df)


# ---------------- Tasks 2 & 3: FactorAnalysis with 2, 3, 4 factors + varimax ----------------
models = {}
fig, axes = plt.subplots(1, 3, figsize=(17, 5.5))
for ax, k in zip(axes, [2, 3, 4]):
    fa = FactorAnalysis(n_components=k, rotation="varimax", random_state=0).fit(Xs)
    models[k] = fa
    L = pd.DataFrame(fa.components_.T, index=questions, columns=[f"F{i+1}" for i in range(k)])
    im = ax.imshow(L, cmap="coolwarm", vmin=-1, vmax=1, aspect="auto")
    ax.set_xticks(range(k)); ax.set_xticklabels(L.columns); ax.set_yticks(range(10));
ax.set_yticklabels(questions)
    for r in range(10):
        for c in range(k):
            ax.text(c, r, f"{L.iloc[r, c]:.2f}", ha="center", va="center", fontsize=8)
    ax.set_title(f"{k} factors (varimax) | avg log-lik = {fa.score(Xs):.3f}")
plt.colorbar(im, ax=axes, fraction=0.02); plt.show()
print("Average log-likelihood by number of factors:", {k: round(float(m.score(Xs)), 3) for k, m in
models.items()})
```

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# ---------------- Task 4: name the factors (3-factor model) ----------------
fa3 = models[3]
L3 = pd.DataFrame(fa3.components_.T, index=questions, columns=["F1", "F2", "F3"])
print("\nLoadings (3 factors):\n", L3.round(2))
print("\nFactor interpretation:")
for f in L3.columns:
    high = L3.index[np.abs(L3[f]) > 0.4].tolist()
    name = Counter(theme_of[q] for q in high).most_common(1)[0][0] if high else "?"
    print(f"  {f}: high-loading questions = {high}  ->  '{name}'")

# ---------------- Task 5: Factor analysis vs PCA ----------------
pca = PCA(n_components=3).fit(Xs)
pca_load = pd.DataFrame(pca.components_.T * np.sqrt(pca.explained_variance_), index=questions,
columns=["PC1", "PC2", "PC3"])
print("\nPCA loadings:\n", pca_load.round(2))
print("PCA variance explained:", np.round(pca.explained_variance_ratio_, 3), "total =",
round(pca.explained_variance_ratio_.sum(), 3))
print("FA noise variance (uniqueness) per question:\n", pd.Series(fa3.noise_variance_,
index=questions).round(3).to_string())

fig, ax = plt.subplots(1, 2, figsize=(12, 5))
for a, M, t in zip(ax, [L3, pca_load], ["Factor analysis (varimax)", "PCA (unrotated)"]):
    im = a.imshow(M, cmap="coolwarm", vmin=-1, vmax=1, aspect="auto")
    a.set_xticks(range(3)); a.set_xticklabels(M.columns); a.set_yticks(range(10));
a.set_yticklabels(questions); a.set_title(t)
    for r in range(10):
        for c in range(3):
            a.text(c, r, f"{M.iloc[r, c]:.2f}", ha="center", va="center", fontsize=8)
plt.tight_layout(); plt.show()
print("""
FA vs PCA: PCA finds components that maximise total variance (a data-compression tool); factor analysis
models
each question as common factors + its own unique noise, so loadings are easier to interpret as hidden
constructs (e.g. 'Compensation'). Rotation (varimax) makes each question load strongly on one factor.""")
```

Output

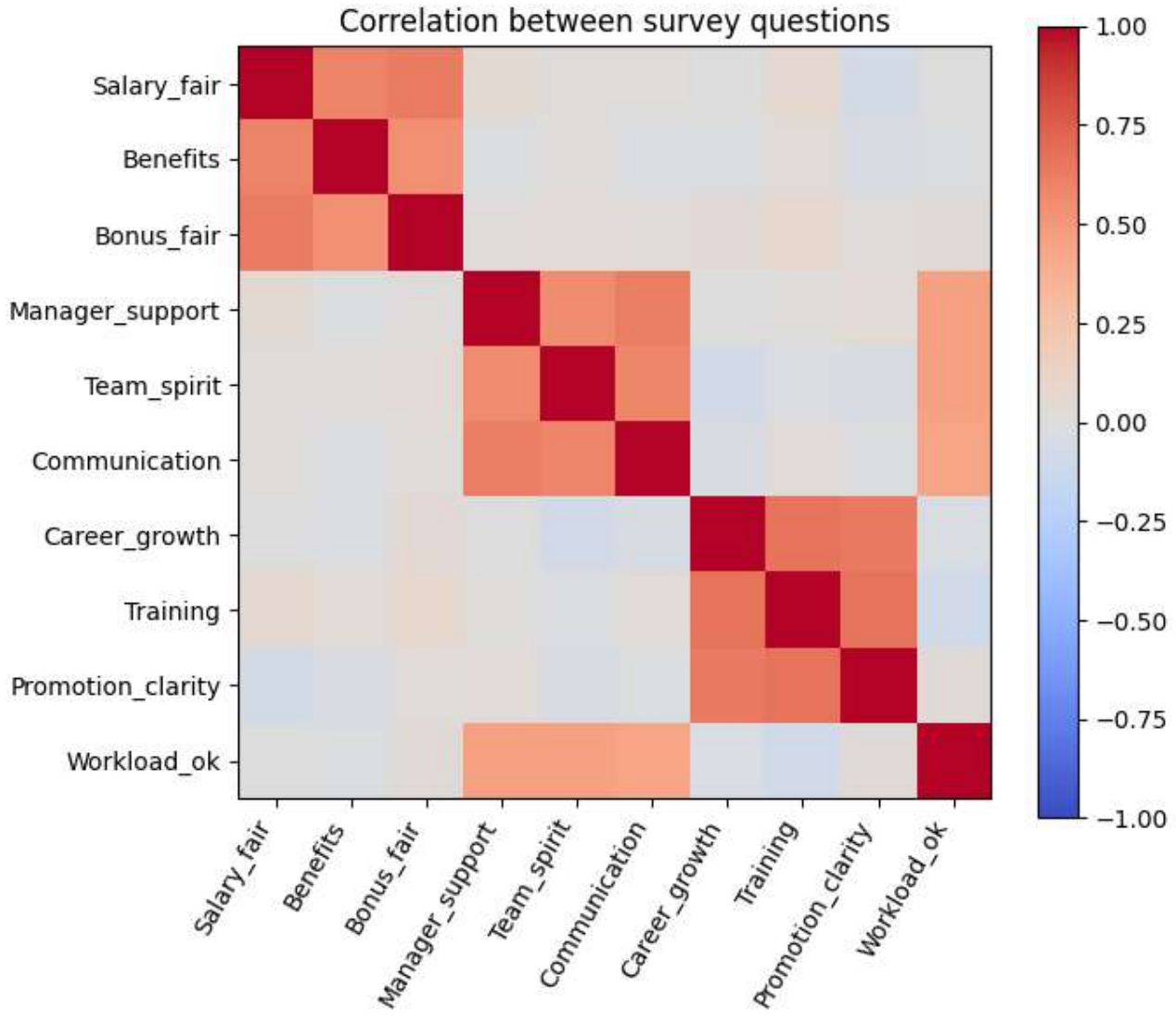
```
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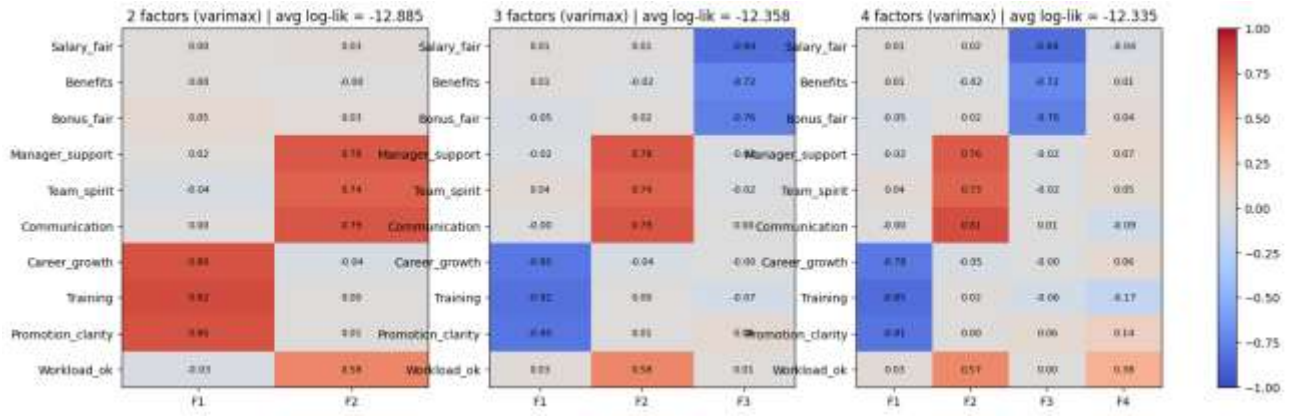
```
   Salary_fair  Benefits  Bonus_fair  Manager_support  Team_spirit  \
0            4         3           3                4            3
1            4         4           4                4            3
2            5         4           4                3            4
3            3         4           3                2            3
4            3         4           4                1            3

   Communication  Career_growth  Training  Promotion_clarity  Workload_ok
0              3              3         4                  4            4
1              3              3         3                  3            3
2              3              2         3                  3            3
3              2              2         3                  3            3
4              1              1         1                  2            2

        Salary_fair  Benefits  Bonus_fair  Manager_support  Team_spirit  \
mean           3.06      3.08        3.05             2.93         2.98
std            0.91      0.87        0.88             0.91         0.95
min            1.00      1.00        1.00             1.00         1.00
max            5.00      5.00        5.00             5.00         5.00

        Communication  Career_growth  Training  Promotion_clarity  Workload_ok
mean             2.94           3.02      3.06               3.04         3.05
std              0.93           0.95      0.89               0.93         0.68
min              1.00           1.00      1.00               1.00         1.00
max              5.00           5.00      5.00               5.00         5.00
```






```
Average log-likelihood by number of factors: {2: -12.885, 3: -12.358, 4: -12.335}

Loadings (3 factors):
                      F1    F2    F3
Salary_fair        0.01  0.01 -0.84
Benefits           0.01 -0.02 -0.72
Bonus_fair        -0.05  0.02 -0.76
Manager_support   -0.02  0.78 -0.02
Team_spirit        0.04  0.74 -0.02
Communication     -0.00  0.79  0.00
Career_growth     -0.80 -0.04 -0.00
Training          -0.62  0.00 -0.07
Promotion_clarity -0.88  0.01  0.06
Workload_ok        0.03  0.58  0.01

Factor interpretation:
  F1: high-loading questions = ['Career_growth', 'Training', 'Promotion_clarity']  ->  'Growth'
  F2: high-loading questions = ['Manager_support', 'Team_spirit', 'Communication', 'Workload_ok']  ->  'Work Culture'
  F3: high-loading questions = ['Salary_fair', 'Benefits', 'Bonus_fair']  ->  'Compensation'

PCA loadings:
                      PC1   PC2   PC3
Salary_fair        0.08  0.26  0.84
Benefits           0.05  0.23  0.80
Bonus_fair         0.08  0.31  0.79
Manager_support    0.81  0.16 -0.18
Team_spirit        0.81  0.09 -0.07
Communication      0.82  0.13 -0.11
Career_growth     -0.20  0.82 -0.24
Training          -0.17  0.85 -0.18
Promotion_clarity -0.15  0.81 -0.30
Workload_ok        0.70  0.08 -0.09
PCA variance explained: [0.258 0.231 0.218] total = 0.706
FA noise variance (uniqueness) per question:
 Salary_fair          0.301
Benefits             0.482
Bonus_fair           0.426
Manager_support      0.398
Team_spirit          0.449
Communication        0.381
Career_growth        0.304
Training             0.310
Promotion_clarity    0.352
Workload_ok          0.663
```

